

T-Route: Channel Routing in the National Water Model

One-Day Hands-On Training Workshop

Date	Wednesday (July 15, 2026)
Time	8:30 AM – 5:00 PM
Format	In-person
Instructors	Md Saiduzzaman & Dr. Bong-Chul Seo

Workshop Overview

T-Route is a tree-based channel routing model that solves one-dimensional routing problems using vector-based river networks. To represent both simple and a little more complex hydraulic conditions, T-Route provides two routing options, the Muskingum-Cunge (MC) and Diffusive Wave (DW) methods. These serve as the primary channel routing module in the NOAA Next Generation Water Resources Modeling (NextGen) Framework.

This one-day, hands-on workshop walks participants through a complete NextGen and T-Route application, including input-data acquisition, rainfall-runoff model calibration, channel routing with both MC and DW routing schemes, streamflow data assimilation, and analysis of results. Working through a guided example, participants will learn the capabilities of T-Route.

Learning Objectives

- Prepare input data for the NextGen workflow.
- Calibrate the NextGen rainfall-runoff model.
- Configure and run T-Route (MC and DW), both within NextGen and standalone.
- Apply streamflow data assimilation in both routing schemes.
- Evaluate T-Route simulation results.

Before You Arrive

- Bring a computer with Docker installed.
- Basic familiarity with Python and the command line is helpful but not required.

Agenda

Wednesday, July 15, 2026

Time	Duration	Session
8:30 – 9:00 AM	30 min	Registration and Welcome
9:00 – 9:30 AM	30 min	Session 1: Introduction to the NextGen Framework and T-Route
9:30 – 10:45 AM	75 min	Session 2: Data Acquisition for NextGen and T-Route
10:45 – 11:00 AM	15 min	Break
11:00 AM – 12:00 PM	60 min	Session 3: Calibration of the NextGen Rainfall-Runoff Model
12:00 – 1:00 PM	60 min	Lunch (provided)
1:00 – 3:00 PM	120 min	Session 4: T-Route Configuration and open-loop Simulation
3:00 – 3:15 PM	15 min	Break
3:15 – 4:45 PM	90 min	Session 5: Streamflow Data Assimilation in T-Route
4:45 – 5:00 PM	15 min	Closing Remarks and Open Discussion

Session Details

Session 1: Introduction to the NextGen Framework and T-Route

- Overview of NOAA's National Water Model and the NextGen Framework.
- T-Route within the NextGen Framework and its routing schemes and implementation (MC, DW, or hybrid).

Session 2: Data Acquisition for NextGen and T-Route

Obtain basic datasets (channel network and meteorologic forcing) for T-Route simulations.

- Process datasets: NextGen hydrofabric river network, channel cross sections, meteorological forcing, and USGS observations.
- Subset the hydrofabric and forcing data for areas of interest.

Session 3: Calibration of the NextGen Rainfall-Runoff Model

Generate channel forcing (lateral flow) data with reduced uncertainty.

- Build the calibration environment including Docker setup and selection of USGS observation site.
- Run the calibration workflow after defining calibration period.
- Review the calibration outputs.

Session 4: T-Route Configuration and open-loop Simulation

Simulate T-Route with multiple routing schemes.

- Configure the T-Route YAML file and define key components.
- Run T-Route within the NextGen Framework
- Run T-Route without NextGen (standalone).
- Evaluate the routing performance between MC and DW.

Session 5: Streamflow Data Assimilation (DA) in T-Route

Incorporate streamflow observations into modeling.

- Enable DA in T-Route configuration.
- Run DA with both MC and DW.
- Analyze the DA results and evaluate them along with the open-loop results.